

ZAIN KHALIL

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EDUCATION

University of Waterloo & Wilfrid Laurier University

Bachelor of Computer Science (BCS), Bachelor of Business Administration (BBA)

Waterloo, ON

Fall 2025 – Present

TECHNICAL SKILLS

Languages & Web: Python, TypeScript, JavaScript, SQL, C, Racket, React, HTML/CSS

Backend & Data: Node.js, Express, FastAPI, REST APIs, PostgreSQL, SQLite, SQLAlchemy, Recharts

AI & Validation: OpenAI API, Pydantic, JSON Schema Validation

Tools: Git, GitHub, pytest, VS Code, Cursor

PROJECTS

SpecGuard: AI Test Intelligence Platform | Python, FastAPI, React, TypeScript, OpenAI API, Pydantic Mar. 2026

- Built an AI-powered QA tool that ingests product specs and generates **schema-validated test suites** with functional tests, edge cases, and negative tests using a **Python/FastAPI** backend and **React** frontend
- Designed a **multi-stage AI pipeline** (parse -> extract -> generate -> validate -> score) with **Pydantic schema enforcement** and automated retry logic on every AI output
- Implemented **structured JSON generation** via OpenAI API, **heuristic coverage scoring** across 4 quality metrics, and a **pytest** suite with 23 passing validation tests

AI Meeting Intelligence Copilot | Python, FastAPI, React, TypeScript, PostgreSQL, pgvector Mar. 2026

- Built a local-first **RAG** system that ingests meeting transcripts, stores chunk embeddings in **pgvector**, and supports **semantic search** over meeting content
- Implemented **structured action-item extraction** using **Ollama**, a **FastAPI** backend, and a **React + TypeScript** frontend for reviewing meetings and updating action items
- Developed an evaluation harness with **retrieval Precision@5**, extraction metrics, and **pytest**-based checks across synthetic transcripts to measure system quality

Personal Expense Tracker (V2) | React, TypeScript, Node.js, PostgreSQL Feb. 2026

- Rebuilt a **Python CLI** expense tracker into a full-stack web application using **React + TypeScript, Node.js/Express**, and **PostgreSQL**
- Designed a **PostgreSQL** data model supporting **expense categorization, time-based queries**, and analytics across **7/30/90-day** views
- Developed interactive dashboards with **Recharts** to visualize spending trends and category breakdowns

EXPERIENCE

Executive Director - OMAM Model United Nations

Aug. 2024 – Mar. 2025

The International School of Elite Education

Cairo, Egypt

- Led end-to-end planning and execution of a fully student-run conference with **~150 delegates**, coordinating delivery across **40+ team heads** and **100+ members**
- Directed budgeting and resource allocation for a **~20,000 CAD** operation, balancing cost constraints with delivery timelines and delegate experience

School President - Student Council

Sep. 2024 – Aug. 2025

The International School of Elite Education

Cairo, Egypt

- Represented and prioritized feedback from **1,500+ students**, translating stakeholder needs into proposals and operational improvements
- Coordinated large-scale events, trips, and school-wide initiatives by managing timelines, communications, and execution across multiple teams

Founder & Coach - Debate and World Scholar's Cup Club

Apr. 2024 – Apr. 2025

The International School of Elite Education

Cairo, Egypt

- Founded and scaled a debate club to **~15 active members**, designing a structured curriculum, weekly materials, and iterative feedback loops
- Prepared students for regional competitions, resulting in **multiple medals and trophies**